Olist E-Commerce
Drop-Off & Customer Journey Analysis

End-to-End Data Analyst + Business Analyst Portfolio Project

Excel / Power Query • SQL Server • Python • Power BI • Business Analysis

Focus: delivery performance, customer satisfaction, operational bottlenecks, seller/category risk, and business recommendations.

1. Business Problem

The project analyzes the Olist Brazilian e-commerce customer journey from order placement through delivery and customer review. The objective is to identify operational bottlenecks, delivery problems, cancellations/unavailable orders, and their relationship with customer satisfaction.

Stakeholders

Business Management, Operations, Logistics, Seller Management, Customer Experience, and Product/Category teams.

Key questions

• Where are problems concentrated in the customer journey?

• Which states, product categories, and sellers show higher delivery risk?

• How does delivery performance relate to customer review scores?

• Which operational areas should management prioritize?

2. Data & Technology

Dataset	Rows
Orders	99,441
Customers	99,441
Order Items	112,650
Products	32,951
Reviews	99,224
Payments	103,886
Sellers	3,095
Geolocation	1,000,163
Category Translation	71

Tools: Excel/Power Query, SQL Server, Python (pandas, NumPy, matplotlib, seaborn), Power BI.

3. Data Quality & Cleaning

Missing delivery dates were investigated against order status rather than blindly imputed or deleted. Delivery timestamps were converted to datetime, relevant numeric fields were validated, and optional review text fields were retained.

Eight orders with status **delivered** were identified without a customer delivery date and treated as data-quality exceptions. Delivery-duration calculations use valid underlying dates; missing dates were not artificially imputed.

4. SQL Analysis

SQL Server was used for data extraction, joins, aggregations, business-focused queries, CTEs, CASE expressions, window functions, and date/string analysis. SQL supported validation and preparation of business metrics for Power BI and Python.

5. Power BI Dashboard Suite

Dashboard	Focus

1. Executive Overview	Sales, orders, customers, AOV, delivery success, trends, categories, delivery status, states
2. Customer & Order Analysis	Payment types, states, delivery time, review scores, AOV, state relationships
3. Delivery Performance	Delivery trends, late performance, state/category/seller delivery risk
4. Customer Satisfaction	Review scores, low-review patterns, delivery impact, category/state dissatisfaction

Interactive slicers were added for year/month and relevant state, payment, delivery-status, and review-score dimensions.

6. Python Deep-Dive Analysis

Python was used as a complementary analytical layer rather than duplicating Power BI. Delivery performance was calculated from actual versus estimated delivery dates.

Performance	Orders	Share
Early	88,649	89.15%
Late	7,827	7.87%
Not Delivered	2,965	2.98%

Category-level late-delivery analysis identified higher-risk product categories. Seller analysis used a minimum threshold of 50 orders to avoid overinterpreting very small sellers.

7. Key Business Findings

Finding	Business meaning
89.15% early	Most completed deliveries occurred before the estimated date.
7.87% late	A meaningful operational segment experienced delays.
2.98% not delivered	Incomplete outcomes should be monitored separately from late deliveries.
Highest observed seller late-delivery rate: 32%	Some sellers show materially higher operational risk.
Early delivery review score: 4.29/5	Early delivery was associated with strong customer satisfaction.
Late delivery review score: 2.57/5	Late delivery was associated with substantially lower customer satisfaction.

Analytical caution: the review-score comparison shows an association, not proof of causation.

8. Root-Cause Analysis Framework

Area	Question
Seller	Are high-delay sellers consistently underperforming or concentrated in particular regions/categories?
Geography	Which customer states have the highest late-delivery rates?
Category	Are particular product categories associated with higher delivery risk?
Operations	Where between purchase, approval, carrier handoff and delivery do delays accumulate?
Customer experience	How does delivery performance correspond with review outcomes?

9. Business Recommendations

• **Prioritize late-delivery monitoring:** track late-delivery rate by state, seller, and category.

• **Target high-risk sellers:** review sellers with persistently high late-delivery percentages and sufficient order volume.

• **Investigate high-risk categories:** examine fulfillment, handling, packaging, and carrier patterns.

• **Protect customer experience:** prioritize proactive communication and exception handling for delayed orders.

• **Separate operational outcomes:** monitor not-delivered, canceled, and unavailable orders separately from late deliveries.

10. KPI Framework

KPI	Purpose
Late Delivery %	Delivery reliability
Early Delivery %	Delivery ahead of estimate
Not Delivered %	Incomplete delivery monitoring
Average Delivery Days	Delivery speed
Average Review Score	Customer satisfaction
Low Review %	Dissatisfaction monitoring
Seller Late Delivery %	Seller operational risk
Category Late Delivery %	Category delivery risk

11. AS-IS / TO-BE

AS-IS

Order placed → payment → seller processing → shipping → delivery → review, with delayed, unavailable, canceled, or incomplete outcomes creating customer-experience risk.

TO-BE

Order monitoring → early exception detection → seller/carrier risk flag → proactive customer communication → resolution tracking → post-delivery satisfaction monitoring.

12. Conclusion

The project combines Data Analyst and Business Analyst capabilities. Excel/Power Query and SQL support preparation and business analysis; Python provides deeper validation; Power BI provides decision-support dashboards; and the BA layer converts findings into root-cause questions, KPIs, and recommendations.

13. Interview-Ready Summary

Analyzed the Olist Brazilian e-commerce customer journey using Excel, SQL Server, Python, and Power BI. Built dashboards covering executive performance, customer/order behavior, delivery performance, and customer satisfaction. Performed delivery-risk analysis by state, category, and seller, and found that early deliveries averaged 4.29/5 while late deliveries averaged 2.57/5. Translated findings into operational KPIs, root-cause questions, and business recommendations.